The Greece-China Connection

Eye-Shaped and Horned Glass Beads from China and Greece

by Theresa Mitsopoulou

This paper was to be presented at the 10th International Conference on Glass hold in Rhodes (1-4 April 2001), but It has been refused "because it was dealing with Chinese and not with Greek glass".

Biliu (li) is one of three words meaning *glass* in Chinese and because glass was made in blue color for protection, in English and French **biliu** meant, finally, blue colour!

The ground in Buddhist heaven is made of coloured glass, which shows how much glass was appreciated. Aristophanes refers to glass as "stone" and this proves, that before being manufactured, it, already, existed in nature as rock and ore. The earliest objects of man-made glass (from about 1500 BC) originate from Mesopotamia and Egypt, but also China, and, it is possible, that China was first in glass manufacturing and deserves as well this priority. Researchers wonder, if the early glass findings of China were imported or local production and if glass was invented independently and parallely in East and West or, did it have a common origin.

In China within remains of Bronze Age kins were found multicoloured beads of a primitive kind of glass, while in later graves (mainly of the 5th c. BC) beads of advanced technology. Their perfection can only be explained by a long tradition of glass-making in China, but, because there are not such records, they, usually, are referred to as "imported from the West". It seems, though, that at least in the 5th c. BC, China, had a well developed glass industry and the belief, that the glass objects were made in China, is strengthened by their shape and use. Many beads from their form and decoration are called "eye-beads", in, usually, oval, round or tube-like shape, from one to three centimeters and a hole for suspension. Similar beads were common in Egypt, Mesopotamia, India, Tibet, even Europe and they are often refered as "Mediterranean" or "Egyptian-Roman". They used to wear them around the neck or the wrist of the hand and on the belt for decoration, but, mainly, as amulets for protection of the individual from evil forces.

The Spectacles of the Naga

Most of them in oval or round shape with painted or in relief decoration, probably, imitate the "spectacles" of the naga, two "dots" the cobra has on the back of the head, which resemble eyes and with symbolic representations of the snake (circles, spirals, wavy lines, meander, etc. like on jewels in general) offered ideal protection. Their usual and predominating colours were blue, white and yellow. In 1978 in the unlooted grave of Yi, marquis of Zeng, in Hubei (the grave was sealed in 432 BC after the burial) about 100 beads were found, because entire gowns were made with glass beads (usually multicoloured and with or without decoration) to accompany the owner to the grave and secure immortality like jade.

In 1977 such a gown was also, found in the grave of a noble lady of te W. Han dynasty consisting of 600 rectangular and circular moulded beads with floral and plant decoration covered with gold leaf. In the 6th c. AD grave of King Muryong and his wife in Korea near Seoul were found hundreds of multicoloured beads in 8 different sizes with hole, which were sewn together to form a unique gown. Beads used to accompany in a string (rosary) monks and laymen during hours of meditation and prayer ("bead" meaning "prayer" - bede). Amber beads not only calmed the nerves, but they also shedded beautiful resine perfume.

The History of Glass

For producing glass were absolutely necessary ceramic kilns and the symbol of the potters and later of the glass manufacturers was the salamander, because it was believed that it can survive high temperatures (its skin diffuses a liquid that helps) and that even settled in the fire.

In China in kilns of the Shang dynasty (1600 - 1027 BC) was discovered a kind of primitive glass, but, officialy, it is refered to glass from the 6th c.BC (like in Greece) with the Eastern Zhou (770 - 475 BC), while it reached high development with the Warring States (475 - 221 BC), with the Western Han (206 BC - 8 AD) and Tang (618 - 907) dynasties.

There is no evidence in written Chinese sources about this glass, if, for instance, it was made in China or if it was imported from the West, ready or as raw material. Texts of the W. Han dynasty clearly mention imported glass from the Roman Empire and, also there is a testimony, that travellers from the West taught in China in the year 435 glass manufacturing. Even the words meaning "glass" in Chinese (like boli) are disputed and explained of Sanskrit origin. However, the shape, the decoration and the use of early Chinese glass items suggest, that they were manufactured in China. It seems certain that also this big discovery was accidental and the Roman Plinius (23-79 AD) tells the story of Phoenicians, who saw sand and natrium to turn into glass in a fire.

In China there is information about early porcelain and bronze, but early use of glass is absent and it remains unknown to which of the two hemispheres its discovery must be attributed.The most ancient examples of Chinese glass are the multicoloured beads (colourless glass was later made), which survived human and natural destructions, because of their small size. Since at least the 5th c. BC there was extended glass manufacturing, but this advanced technology could be only explained by a long tradition of more than 2,000 years for which there is no evidence and the possibility, that this glass was imported from the West is, therefore, justified.

In Egypt, glass appears directly in an advanced form and it is probable that Pharaohs, like

Touthmoses III, brought back with them after victorious campaigns in Asia qualified workers, that already knew about glass manufacturing.

With the Alchemy of the Taoists inorganic materials of nature mixed together and under high temperatures produce now glass, which will mainly imitate jade and it will inherit its potential to grant immortality.

Solid glass beads, easy to be manufactured, will be followed by concave items, like cups and bottles and by the core technique. Mainly, since 1950 were excavated in many provinces (like in Hubei, Hunan, Henan and Sichuan) early glass objects, e.gr. arrows, items for the decoration of spades, swords and daggers, belt buckles, bracelets, idols, etc.

Glass will copy shapes of Chinese ceramics, which do not allow any doubt about the country of production of the glass and China was anyway first in manufacturing glass with barium and lead. Since the 14th c. Poshan in Northern China will develop into a big center of glass, because quartz was easy to be found very near. It was melted in kilns and then in rods was sent to Peking, where after second melting were manufactured different items in many colours.

The center in the South will be Canton, while since 1800 during the Qing dynasty glass masterpieces in 30 colours will be as well produced in the Forbidden city of Peking. From the W. Han dynasty (about the same time in Greece) the fused glass will allow mass production and will diminish the cost. Producing glass by blowing air into the glass has been recorded since the 1st c. BC, but there is evidence that the technique was known earlier. The man-made glass was either stuck when warm to the iron rod or was poured liquid or in powder in moulds before it was put in the kiln. The natural glass, like rock crystal, was worked by hand and it was more difficult to give to the bead, for instance, the round shape than to make it in a mould.

Glass was known to the ancient Greeks, but they did not make extended use up to the Hellenistic period.
Beads of rock crystal were found in Mycenae, but bigger objects, (like the cup of Kakovatos) were, probably, imported, at least the material, because in the Greek nature (Crete, Taygetos, etc.) there is no rock crystal in such big size. On the contrary there was plenty of obsidian (petrified lava), especially on the island of Milos and it was used for weapons and tools.

The Decoration of the Eye-shaped Bead: The Eye of Knowledge

Precious and semiprecious stones, diamonds, rubies, lapis lazuli, jade, agate, coral, in eye-like shape (oval, almond), will decorate earrings and rings, horizontally or vertically, in China and elsewhere, up to today. The shape will symbolize the snake (the spectacles of the cobra), the best defense against Evil.
The snake-protector ate the rats, who spread the cholera, illness which threatened mankind with extermination. Protection needed the sensitive and important human eyes and eye glasses.

Today, the frame and the "arms", have symbolic snake representation (rombuses, zig-zag, meander, "metops and triglyphs", "hearts" - head of the cobra, applique snakes in relief and "eyes") like on Oackley and Smith eye glasses or imitate snake skin in fish scale design, while the "eagle" of the Armani glasses is in fact a winged snake (inside the Acropolis Museum one of four big stone

snakes, the "Triton", has wings).

Similar patterns will decorate fabrics and belts, while athletic shoes (Nike, Adidas, Intern and Star) are, also, decorated with "eyes" and who knows what kind of mechanism preserved in a world scale these superstitious designs, which have copied the same up to now.

In the movie "Magtub, the Law of the desert" Omar Sarif as emir of a desert tribe in Maroco wears a big vertical "eye" of gold and precious stone on his headdress (the eye of knowledge).

In Malta, following the ancient Chinese, Greek, and Etruscan tradition, painted "eyes" are indispensable on the prow of boats and ships. Besides, in Malta "eyes" in relief are hung on the wall of the rooms and big "eyes" are built on the façade houses.

With this "eye" ("an eye for an eye") the Evil eye will be knocked down. Certain beads have conical decoration in relief, usually, 6 cones with painted or incised concetric circles (usually 5 parallel circles), which resemble the "breasted" ewers of Santorini, but, also Chinese vases of the same period or even older ones.

These conical protuberances imitate, most probably, the "horn" on the head of certain cobras, while the so-called "heart" (emblem of the Medical Center of Athens) symbolized the head of the cobra and averted Evil like the "horn". The spirals of the Ionic style capital are formed by coiled snakes (not by the horns of the ram) and in the Museum of Siphnos two snakes are to be clearly distinguished on a votive capital. The protuberance-eye in the center of the spiral was painted blue (like on the capital of the Athenian Agora), while the "hole" - eye of the Aeolian capital was filled with lapis lazuli.

The Heavenly Eye-beads of Tibet

Their name is due to the belief that Gods left them fall from the skies. Most of them are tube-like, elongated beads (few are sphaerical), in black or red colour and, usually, made of clay or agate. Agate was one of the stones (like lapis lazuli, amber, jade, etc.), that could keep Evil away and a kind of agate is called "eye-stone"- opthalmolithos, because from its nature has in the center an almond-like or round "eye", something that explains why most heavenly beads were made with this stone. In Tibet they were known before Buddhism and they were worn as amulets with supernatural and magical forces, warding off Evil, bringing health and wealth, good fortune and peace.

They strengthened the metabolism (it is believed even today that they help lose weight) and the blood circulation. They have, usually, from two to nine "eyes" and their value depends on the number of eyes.
The ancient, original ones, are considered to be about 1300 years old and Museums offer astronomical prices in order to obtain them. The Tibetans appreciate and respect very much these beads, because they believe that they originate from living organisms, when Tibet was, millions of years ago, in the bottom of the sea and they still present them to the Buddha.

The Beads of the Vikings: The Compass and the Marbles

The Vikings, like the Chinese were fond of lucky games and they used glass, multicolored beads with patterns against Evil (spirals, wave lines, circles, etc.) as dice. Unrivaled navigators, that went to

America 500 years before Colombus, for sure, like the Phoenitians, they had to travel for their commerce up to China. They named their ships "dracar" from the monsters that decorated their praw, like on the Dragon boats of China and this way is explained their advanced astronomical knowledge, absolutely necessary for the navigation. It must be therefore not true that they ignored the compass and on a 11th c. wooden tablet from Dublin an incised representation of a ship has above the central mast a fish. It is known that in the steppes and China the compass had at first the shape of a fish and as a compass must be explained the fish on the praw of Greek neolithic "frying pans" (in this case the use of the compass at sea is thousands of years earlier).

The raw material for the beads the Vikings did not manufacture themselves, but for producing glass they imported glass dice from W. Europe (N. Italy) and they also recycled broken glass objects. Well known glass producing center (besides Ribe in Jutland and Paviken in Gotland) was the ancient city Birka in Sweden, commercial center of Scandinavian and Baltic countries. Glass was melted in funnels, it came out in strips and yet soft was wound around iron rods, while after second melting it was poured into moulds. With similar glass beads like those of the lucky games of the Vikings used to play on the street Greek kids.

In Steven Spillberg's movie "The Empire of the Sun" boys play marbles in a Japanese concentration camp. Their name in Greek was "yialinakia", "boloi" and "gazes" - "marbles", small glass beads and one big, called "mana" - mother (the poor kids had clay beads). In small pits on the ground they used to throw in from far as many beads as possible, a kind of "billiards" (the Italian word "biglia" is, possibly the Chinese word "biliu(li)", which, also means "glass", while Dr. R. Brill of the Corning Museum of Glass is of the opinion that the Greek word "bolos" - bead is the Chinese one "boli" - glass).

Renowned are as well the glass beads of the Etruscans, which like the "smile", the ceramics, the antefix, the bronze vessels, etc., connect this enigmatic people with China. Early beads of Japan and early glass of Korea also originate in China.

The Blue Color of the Glass

A third Chinese word for "glass" is "liuli", which exactly means a blue-green glaze, from a plant of India, indigo. Indigo in Chinese is called "lan" and it is possibly the syllable "lan-" of the Greek word "galanos"-blue and it was used ("loulaki" in Greek) as a whitening in washing, while "loulaki" means the color blue. It was the first color that did not exist in nature and had to be manufactured. In the peninsula of Sinai the Egyptians since 4,000 BC mined lapis lazuli (turquoise) at Malkat, which will be a second name to lapis lazuli. The name "turquoise" is due to Turkestan, where this stone is still mined and its best variation, today, comes from Ispahan in Iran.

The plant "indigo", lapis lazuli in powder, bronze and cobalt from Afghanistan were used to make the blue color. The color blue the Chinese distinguish in blue of the sea, of the lake, of the sky, of the peacock and in cobalt blue. To this color will be attributed magical powers and it was believed that the length of its wave and its radiation were able to keep Evil away.

On the Santorini murals heads of children are partially painted, turquoise-blue, something also common in China.

In Egypt, today, in the Arab villages of Israel, in Northern Africa, the walls of houses are painted blue, so that mosquitoes and flies will not enter. In ancient Greek architecture one of four primary colours was turquoise-blue and in that colour are painted the doors and windows of the Cyclades.

The same blue colour was given to the glass for protection, not to break, and, it seems, that the name of the glass "biliu(li)", finally, meant in English and French not only the glass, but also the colour "blue".
Famous, today, is the blue glass of the Arabs, but about 900 Arabs had settled in Southern China (Canton), where besides Astronomy, Mathematics, Shipbuilding and Navigation they, possibly, learned how to make the blue glass and yet, finally, better than their masters.

Important center of fused glass was founded in Malta, when the Arabs settled, in 870. The traditional location, today, about 10 kil. from the capital Valletta, is called "Ta Quali" ("ta" means "to"), but nobody has thought to link the word "quali" to the Greek word "ualos" - glass. In modern Greek a - y was added (yuali) in the beginning of the word like in uios-yuios, while in Arabic a - q (the -u is pronounced -u in tripa-troupa). Today, in Maltese is used a different word for "glass", but the word "ualos" survived on the small hill of Malta and offers its testimony to the history of glass. The Arab word "quali", probably, is a Chinese loan and, moreover, the Greek word "ualos" was in use in China, at the time the Arabs learned there how to make glass. The Latin word "vitrum", which meant "glass", originally was the name of a plant (with its leaves a dye was made similar to the indigo, called "pastel"). The English name of the plant is "woad" and the Bretons, invincible warriors, used it to paint blue their faces before the battle.

Mirrors and the Winthrop Mirror

Written sources do not agree if the handsome youth Narcissus saw his figure in the water of a fountain or of a river, it seems, certain though, that man saw his own image for the first time reflected on the still water surface. The early Emperors of China used as a mirror big clay and bronze vessels with water. In Greece, since Mycenean times used hand mirrors made of shining metallic surface like of silver and bronze with wood or ivory handle, which, according to Mythology, were invented and forged by Hephaestus.

The usual diameter of the disc was 15-20 cent. and on the outer side was decorated with designs, painted, incised or in relief. It is believed, that the inner front surface was very shiny and reflected the image, but it is also possible that this side had, already at that time, a second disc of glass coated with silver and tin like, today, on the famous hand mirrors of popular art of Ioannina, where the glass disc is applied on the silver one and is kept in place by metallic hooks.

The possibility that metallic mirrors also had glass disc is strengthened by the famours Winthrop mirror, a rare item of about 400 BC (diameter 12,3 cent.), today, in the Fogg Art Museum, MA. It was found in a royal grave in Hunan province, where once flourished the mighty and rich Chu Kingdom. The back surface of the bronze disc is decorated with applied eye-shaped glass beads. Some of them surround a ring of jade in the center with "rope" decoration and most of them are surrounded by a jade ring. Similar rings are known from elsewhere and they even have head and tail of a snake like the dragon-pendant of the Warring States period in the Asian Art Museum of San Francisco (Coll. Alan Feen) and do not allow any doubt about the symbolism of the rings.

As regards the combination of glass and jade on the same object is unique. Besides a big bead, which is surrounded by 7 smaller ones and all of them together form a "rosette" - lotus flower in the center, the disc is decorated with 30 more multicolore beads in two circles of 18 and 12. Seen that the eye-shaped bead had the power to keep Evil away and taking into consideration the fact that it decorated glass artefacts like belt buckles, cups, etc., so that they will not break, it seems probable that this mirror had attached a second glass disc.

The mythical Chinese Yellow Emperor had ordered his craftsmen to manufacture 12 mirrors for him, one for every month of the year, but there is no information about the material, that they were made. Mirrors reflected as well the social level of the owner and they were indispensable in the dowery of the bride. In the grave of a concubine of the last king of the Shang dynasty mirrors were found for the first time.

The Palace Museum of Taiwan numbers about 150 ancient mirrors, most of them from Continental China. Mainly bronze hand mirrors, but also for the table, mostly circular but also square (Heaven and Earth) have the back side beautifully decorated and many have in the center of the back a hook attached for a string to be hung. It is believed, that in China the first glass mirrors were manufactured towards the end of the Qing dynasty (1694-1911) and in Europe about 1900. In Versailles mirrors on the wall (Galerie des glaces) contributed to the splendour of the Palace.

Roman and Venetian Glass

Important glass production in China is documented in the W. Han dynasty, when at the same time centers of glass production flourish in Europe and in the Eastern Mediterranean and Persia. Glass objects of this period, found in China, after chemical analysis seem to be rather imported from the West and products of workshops of the Romans and of the Sassanids, which came to China mainly from the sea, but, also, over the land Silk Route.

Rome and after that Byzantium the Chinese will call "Da Qin", "the big kingdom of Qin" (Qin being the richest and most powerful of nine kingdoms, whose king, the famous Qin Shi Huang "united" China by conquering all other kingdoms and took the name of the first "Emperor").

The W. Han dynasty was in constant communication with Rome and the Roman glass, superior to any other earlier glass, will be imported into China and will be exchanged with gold and silk. The Roman legions will bring the Roman glass to Germany, France, Netherlands and Britain.

The mass production of glass was delayed because high temperatures (1800C) were required for melting the quartz. Later, it was melted in lower temperatures (today only 1000C) by adding potash or sodium carbonate, and the glass was first put then in an oven of 400C to cool off, gradually, and to avoid breaking.

For creating a glass center, besides sand, was necessary extended woods for the kiln. Venice, whose the strong relations with China are symbolized by Marco Polo, disposed not only plenty of sand from the salt lakes, but also a lot of timber and it soon became an important glass center (Murano). In spite of the fact that the secrets of glass production were well protected (the penalty was death for those, that could not keep their mouth shut), it will be spread from Venice to Bohemia, and, later Czeckoslovakia, England and America will also rise into famous glass centers.

But the shape and decoration of the glass products of Venice bring them in contact with China. Glass vessels have stems and handles in the shape of snakes, horizontal lines around the neck, where, usually, the snake was symbolized, herring and zig-zag decoration, rombuses, meander, spirals and the head of the cobra, which is explained as "heart" of "lotus leaf".
Vessel in the shape of lotus flower from Persia in the Corning Museum of Glass brings again China in mind as shapes of vessels inspired from the lotus flower had long tradition in China in the Palace and the Temple, like the renowned gold bowl of the Kypselids, today in Boston in the Museum of Fine Arts.
The indirect approach of the history of glass regards the shape and the decoration of certain objects like the small flasc-pendant of 1,500 BC from Mesopotamia, usual shape of Chinese vessels' (a similar one was found on Santorini). Also glass of oil lamps, in the form of a gourd, resembles the "twin vases" of China (known also from Troy) in two horizontal or vertical levels, which symbolize the Yin-Yang (Positive-Negative).

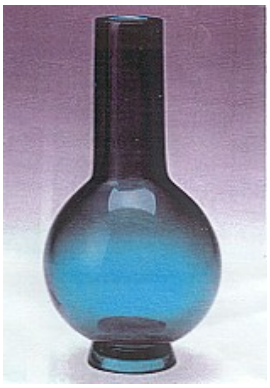
It is possible, that the colored glass of Byzantine and Gothic Architecture, the glass of the windows of houses, the murals, icons and floors in mosaic have been, originally, achieved in China.

Glass Museum in Rhodes

Today, besides the famous Museum of Glass at Corning, N.Y., where a long glass industry, and the recently inaugurated Grand Crystal Museum of Taiwan, Glass Museums have Germany, Italy and Czechoslovakia.

Greece has unique items of natural and man-made glass, which could form the nucleus of a Glass Museum, probably in Rhodes, which had important glass production in the Hellenistic period. Moreover, Rhodes is the island, which hosted in the year 2001 the 10th International Glass Conference, and where lovers of this wonderful material came from all over the World.

Eye-Shaped and Horned Glass Beads from Greece and China



Quing (ca 1800) blue-green vase in the Asian Art Museum of Sab Francisco. *Biliu* is one of three words meaning *glass* in Chinese. This color was used to prevent breaking. In English of course the word means blue color.

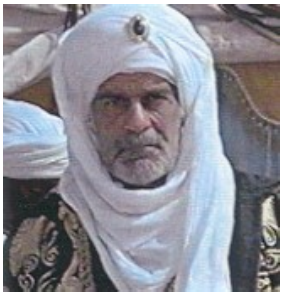
Modern eye-beads from Turkey in the traditional protective colors of blue, yellow and white

The eye-shaped beads imitate the 'spectacles' on the back of the head of the cobras.

W. Han dynasty ring with rubies from Xinjiang. Rings with vertical or horizontal eyes are still made even now.

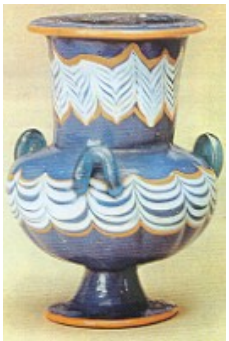
Hellenistic glass beat from the Treasure of Kechreae, with eyes, from the museum of Isthmia.

Egyptian glass cup (800 BC) with glass ring eyes abd a serpent handle.



The "Eye of Knowledge"
Omar Sharif as the Emir of the Beni Zair

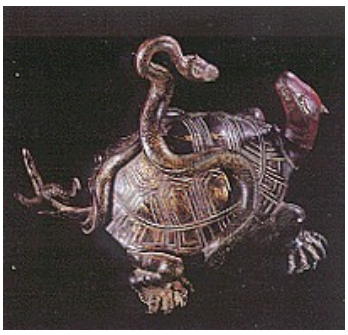
Glass beads of the Warring States in the Hunan Museum



Egyptian glass vase (ca 1500 BC) in the traditional protective colors of blue, yellow and white.

Tube-like and two round beads of the Warring States in the Museum of Hubei

Funeral marble vase of the 4th c. BC with handles formed by dragons from the Athens Archaeological Museum



Bronze turtle with horned viper of the Ming Dynasty(1368-1644) in the Museum of Hubei

Stem in shape of a dragon. Venitian 15th c. glass goblet from the Corning Museum of Glass in N.Y.

'Heavenly beads' are offered bt the believers to Buddha

Ten beads of the Warring States from Hubei and Henan.
One is a horned bead.

The horned conical bead has the horns decorated with parallel lines

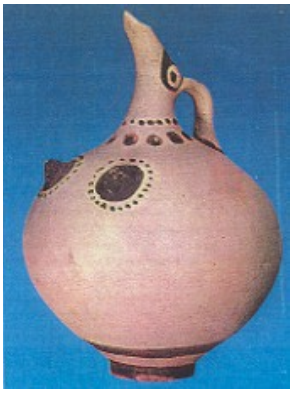
Round beads (and a conical one) of the 5th and the 4th c.BC in the British Museum. They contain barium and were probably manufactured in China.

Glass bead of the Warring States period (475-221 BC) in the Asian Art Museum of San Francisco

Bead in the Museum of Troy

'Eye-stones' with round and oval eyes.

Shang (1600-1027 BC) bronze vessel in the Museum of Hubei with conical horns with concentric circles



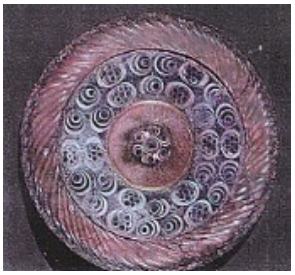
'Breasted' ewer of the 17th c. BC from Santorini. Around the nipple, dots symbolizing the snake and on the neck an 'eye' to prevent breaking.

Modern tatoo (eye) on the arm. Many people wear bracelets in the shape of snakes

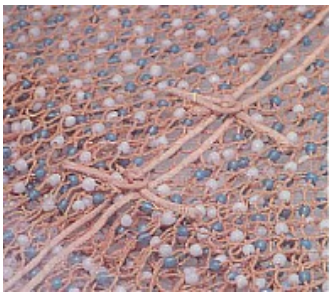
Dancer in a nightclub in Taiwan with hearts in a circle around the nipple of the breast.

Clay Huaxtecan (about 1400) figure in the Museum of Anthropology of Mexico city with symbolic snake decoration and 'eyes' on the hand and arms. Around the nipple of the 'horned' conical breats are small dots.

Glass beads which once formed a dress from the grave of King Muryong near Seoul (6th c)

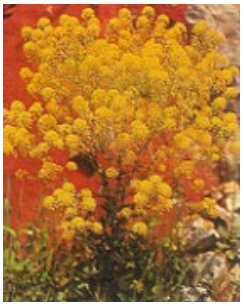


The bronze Winthrop mirror in the Fogg Art Museum, MA with 'revolving eyes' decoration to prevent breaking. Most probably the inner surface had a real glass mirror applied like on the traditional bronze mirros of the Greek Folk Art of Ioannina



Peking Opera actor's dress of glass beads

Curtain with 'eyes' in a nightlcub in Peking



The plant 'isatis' (indigo, lan, vitrum, woad) that produced the blue color

Gold medal of Constantine the Great. On his forhead an eye like the headress of the Emir of Morocco.

Detail of a 6th Century ivory plaque in the Louvre. The horse of the Byzantine Emperor has an 'eye' on the forehead and on the necklace as well.

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